NAND TO TETRIS

NAME: - Shresth Priyadarshi

ROLL NO.: - 2501410026

Project 1

NOT Gate

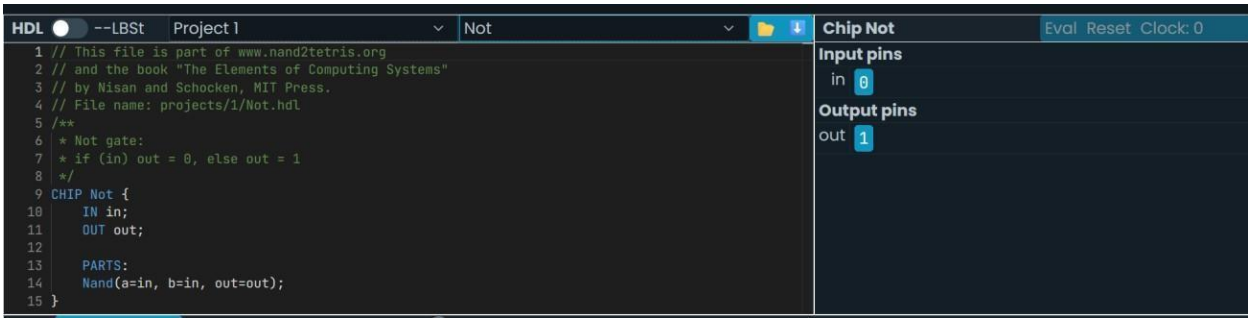
Code

```
CHIP Not {
    IN in;
    OUT out;

    PARTS:
    Nand(a=in, b=in, out=out);
}
```

NAND TO TETRIS

Screenshot



AND Gate

Code

```
CHIP And {

   IN a, b;

   OUT out;

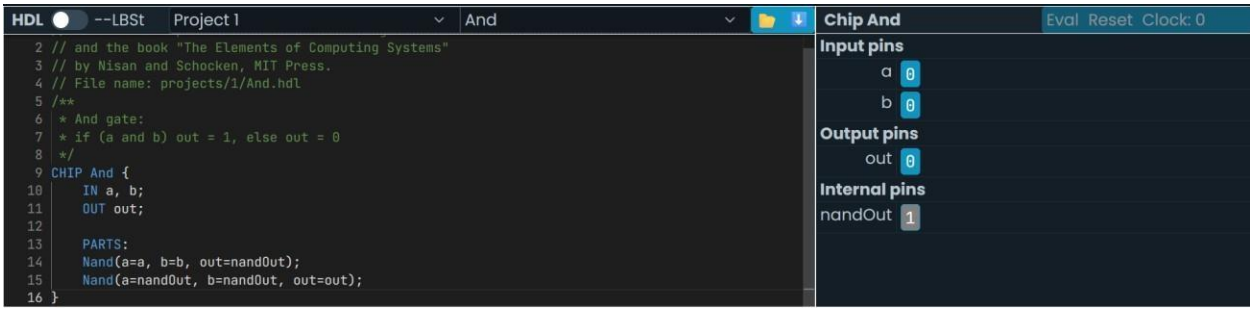

   PARTS:

   Nand(a=a, b=b, out=nandOut);

   Nand(a=nandOut, b=nandOut, out=out);

}
```

Screenshot



NAND TO TETRIS

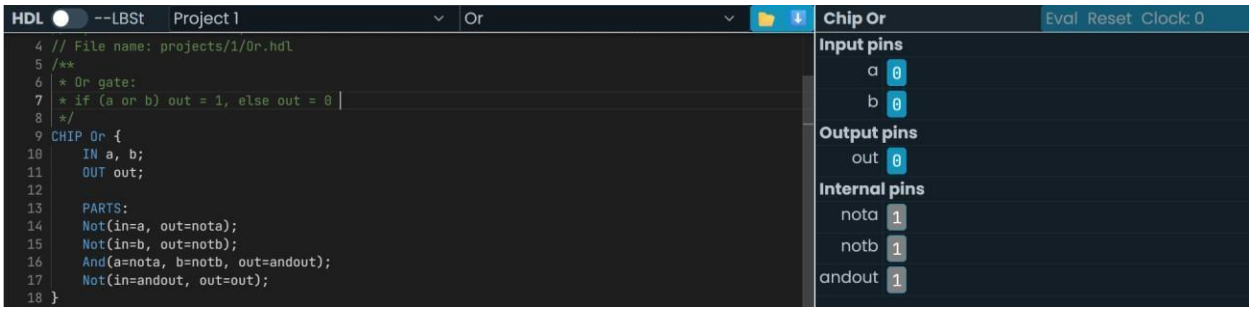
OR Gate

Code

```
CHIP Or {
    IN a, b;
    OUT out;

    PARTS:
    Not(in=a, out=nota);
    Not(in=b, out=notb);
    And(a=nota, b=notb, out=andout);
    Not(in=andout, out=out);
}
```

Screenshot



XOR Gate

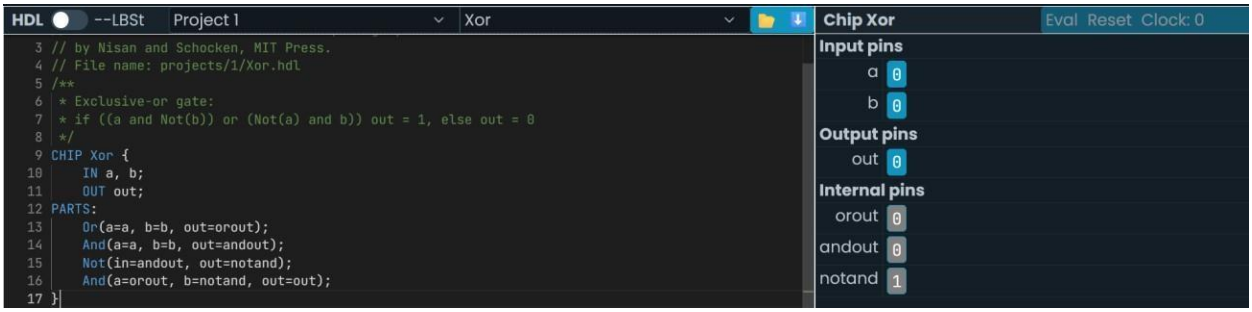
Code

```
CHIP Xor {
```

NAND TO TETRIS

```
    IN a, b;
    OUT out;
PARTS:
    Or(a=a, b=b, out=orout);
    And(a=a, b=b, out=andout);
    Not(in=andout, out=notand);
    And(a=orout, b=notand, out=out);
}
```

Screenshot



NAND TO TETRIS

MUX Gate

Code

```
CHIP Mux {

   IN a, b, sel;     OUT out;

PARTS:

   Not(in=sel, out=notsel);

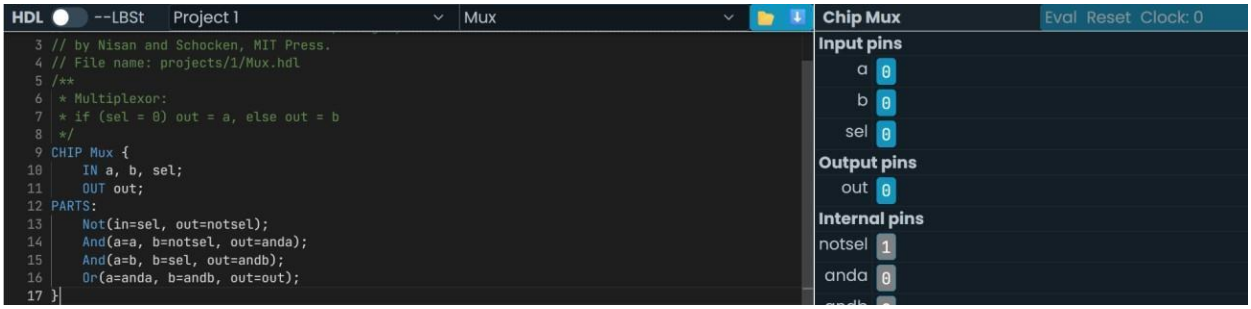
   And(a=a, b=notsel, out=anda);

   And(a=b, b=sel, out=andb);

   Or(a=anda, b=andb, out=out);

}
```

Screenshot



NAND TO TETRIS

Dmux Gate

Code

```
CHIP DMux {

   IN in, sel;     OUT a, b;

PARTS:

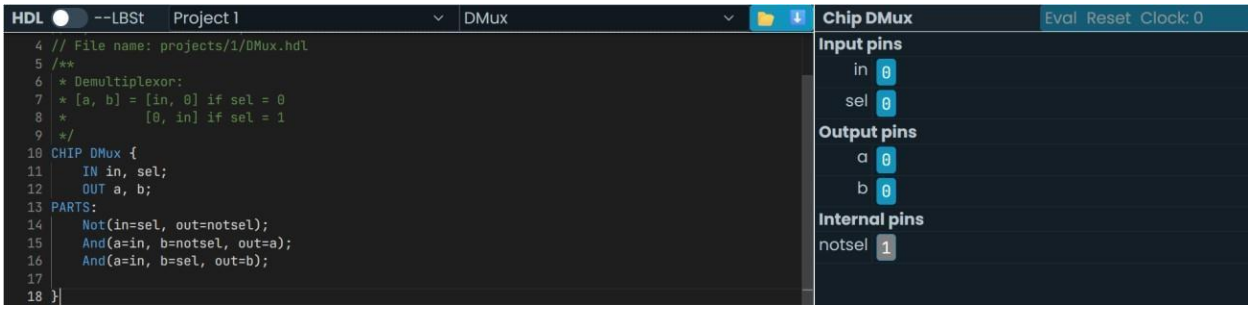
   Not(in=sel, out=notsel);

   And(a=in, b=notsel, out=a);

   And(a=in, b=sel, out=b);


}
```

Screenshot



Not 16 Gate

Code

```
CHIP Not16 {

   IN in[16];

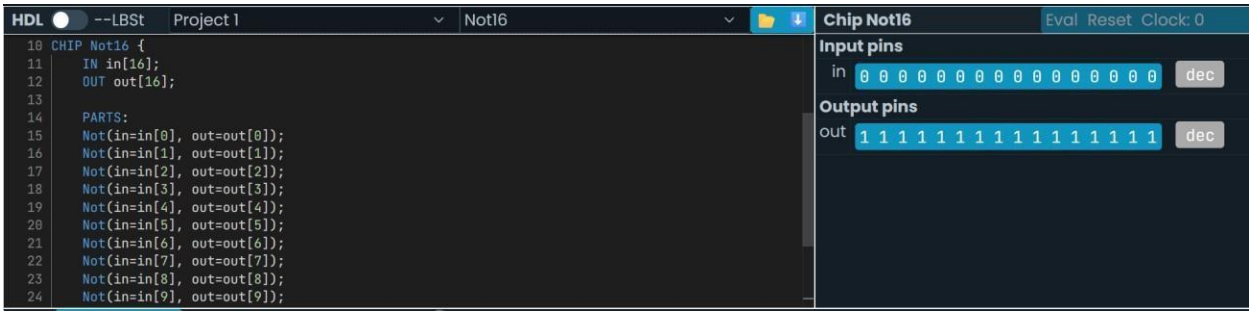
   OUT out[16];


   PARTS:
```

NAND TO TETRIS

```
    Not(in=in[0], out=out[0]);
    Not(in=in[1], out=out[1]);
    Not(in=in[2], out=out[2]);
    Not(in=in[3], out=out[3]);
    Not(in=in[4], out=out[4]);
    Not(in=in[5], out=out[5]);
    Not(in=in[6], out=out[6]);
    Not(in=in[7], out=out[7]);
    Not(in=in[8], out=out[8]);
    Not(in=in[9], out=out[9]);
    Not(in=in[10], out=out[10]);
    Not(in=in[11], out=out[11]);
    Not(in=in[12], out=out[12]);
    Not(in=in[13], out=out[13]);
    Not(in=in[14], out=out[14]);
    Not(in=in[15], out=out[15]);
}
```

Screenshot



AND16 Gate

Code

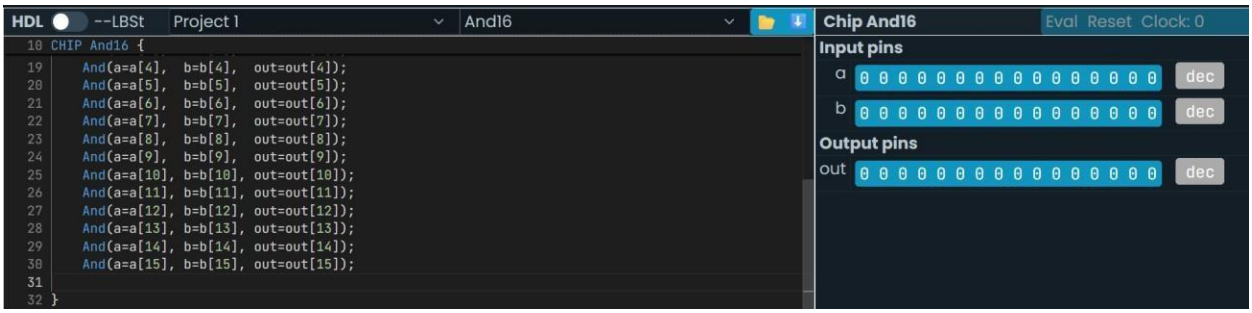
```
CHIP And16 {
    IN a[16], b[16];
    OUT out[16];

    PARTS:
    And(a=a[0], b=b[0], out=out[0]);
```

NAND TO TETRIS

```
  And(a=a[1], b=b[1], out=out[1]);
  And(a=a[2], b=b[2], out=out[2]);
  And(a=a[3], b=b[3], out=out[3]);
  And(a=a[4], b=b[4], out=out[4]);
  And(a=a[5], b=b[5], out=out[5]);
  And(a=a[6], b=b[6], out=out[6]);
  And(a=a[7], b=b[7], out=out[7]);
  And(a=a[8], b=b[8], out=out[8]);
  And(a=a[9], b=b[9], out=out[9]);
  And(a=a[10], b=b[10], out=out[10]);
  And(a=a[11], b=b[11], out=out[11]);
  And(a=a[12], b=b[12], out=out[12]);
  And(a=a[13], b=b[13], out=out[13]);
  And(a=a[14], b=b[14], out=out[14]);
  And(a=a[15], b=b[15], out=out[15]);
}
```

Screenshot



OR16 Gate

Code

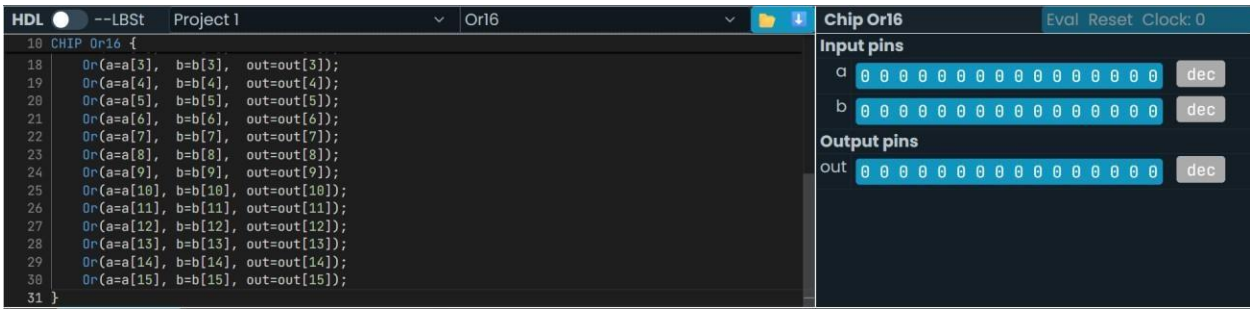
```
CHIP Or16 {
  IN a[16], b[16];
  OUT out[16];

  PARTS:
   Or(a=a[0], b=b[0], out=out[0]);
  Or(a=a[1], b=b[1], out=out[1]);
```

NAND TO TETRIS

```
    Or(a=a[2], b=b[2], out=out[2]);
    Or(a=a[3], b=b[3], out=out[3]);
    Or(a=a[4], b=b[4], out=out[4]);
    Or(a=a[5], b=b[5], out=out[5]);
    Or(a=a[6], b=b[6], out=out[6]);
    Or(a=a[7], b=b[7], out=out[7]);
    Or(a=a[8], b=b[8], out=out[8]);
    Or(a=a[9], b=b[9], out=out[9]);
    Or(a=a[10], b=b[10], out=out[10]);
    Or(a=a[11], b=b[11], out=out[11]);
    Or(a=a[12], b=b[12], out=out[12]);
    Or(a=a[13], b=b[13], out=out[13]);
    Or(a=a[14], b=b[14], out=out[14]);
    Or(a=a[15], b=b[15], out=out[15]);
}
```

Screenshot



MUX16 Gate

Code

```
CHIP Mux16 {

    IN a[16], b[16], sel;

    OUT out[16];


    PARTS:

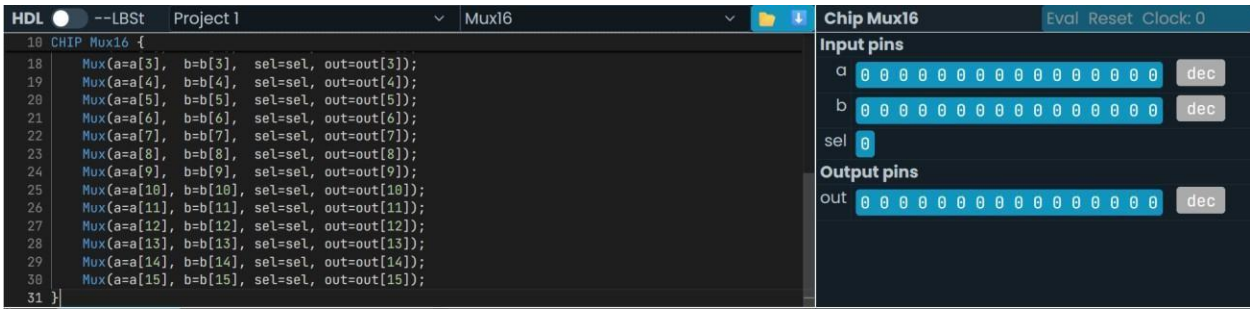
    Mux(a=a[0], b=b[0], sel=sel, out=out[0]);

    Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
```

NAND TO TETRIS

```
    Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
    Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
    Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
    Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
    Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
    Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
    Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
    Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
    Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
    Mux(a=a[11], b=b[11], sel=sel, out=out[11]);
    Mux(a=a[12], b=b[12], sel=sel, out=out[12]);
    Mux(a=a[13], b=b[13], sel=sel, out=out[13]);
    Mux(a=a[14], b=b[14], sel=sel, out=out[14]);
    Mux(a=a[15], b=b[15], sel=sel, out=out[15]);
}
```

Screenshot



MUX4WAY16 Gate

Code

```
CHIP Mux4Way16 {

    IN a[16], b[16], c[16], d[16], sel[2];

    OUT out[16];


    PARTS:

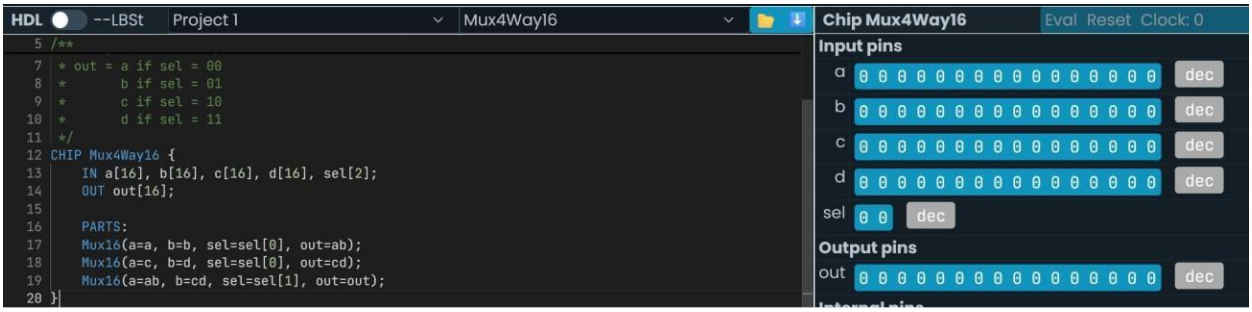
    Mux16(a=a, b=b, sel=sel[0], out=ab);
```

NAND TO TETRIS

```
    Mux16(a=c, b=d, sel=sel[0], out=cd);

    Mux16(a=ab, b=cd, sel=sel[1], out=out);
}
```

Screenshot



NAND TO TETRIS

MUX8WAY16 Gate

Code

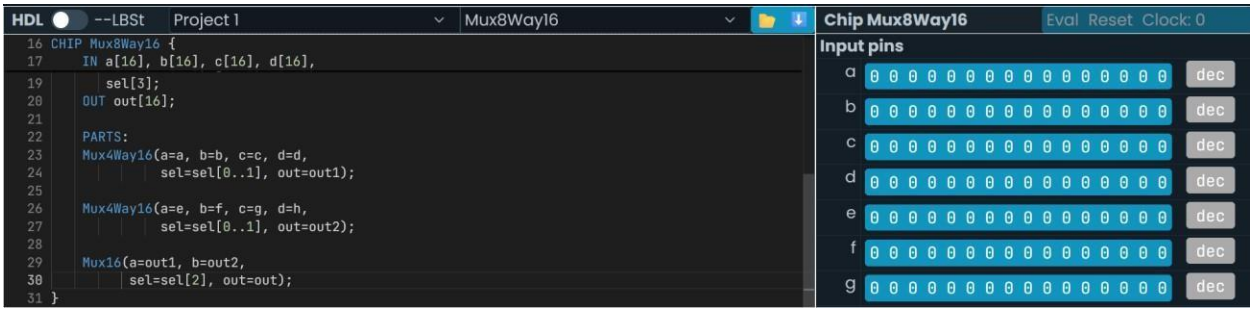
```
CHIP Mux8Way16 {
    IN a[16], b[16], c[16], d[16],       e[16], f[16], g[16], h[16],
sel[3];
    OUT out[16];

    PARTS:
    Mux4Way16(a=a, b=b, c=c, d=d,             sel=sel[0..1], out=out1);

    Mux4Way16(a=e, b=f, c=g, d=h,             sel=sel[0..1], out=out2);

    Mux16(a=out1, b=out2,          sel=sel[2], out=out);
}
```

Screenshot



NAND TO TETRIS

DMUX4WAY Gate

Code

```
CHIP DMux4Way {

   IN in, sel[2];

   OUT a, b, c, d;


   PARTS:

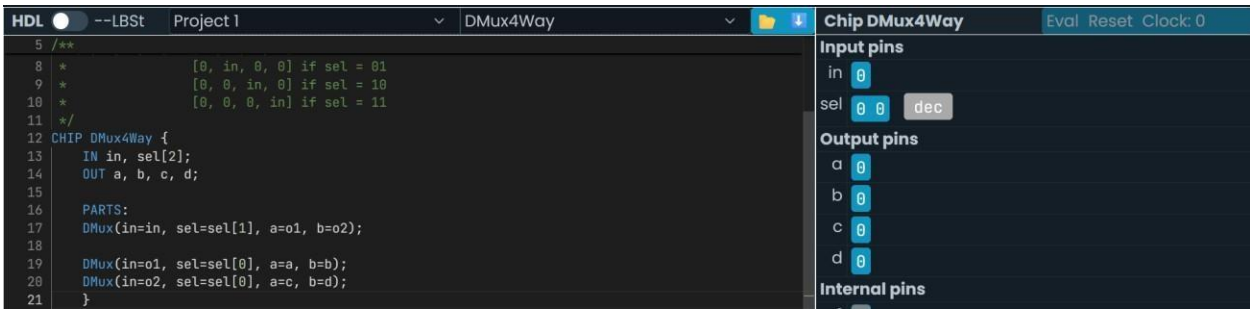
   DMux(in=in, sel=sel[1], a=o1, b=o2);


   DMux(in=o1, sel=sel[0], a=a, b=b);

   DMux(in=o2, sel=sel[0], a=c, b=d);

   }
```

Screenshot



NAND TO TETRIS

DMUX8WAY Gate

Code

```
CHIP DMux8Way {

   IN in, sel[3];

   OUT a, b, c, d, e, f, g, h;


   PARTS:

   DMux(in=in, sel=sel[2], a=topGroup, b=bottomGroup);

   DMux4Way(in=topGroup,    sel=sel[0..1], a=a, b=b, c=c, d=d);    DMux4Way(in=bottomGroup, sel=sel[0..1], a=e, b=f, c=g, d=h);    }
```

Screenshot

